ARC-SPEC-2026-EDU-09: Universal Systems Transduction & Interdisciplinary Field Mapping

A Mathematical Framework for Cross-Domain Admissibility in Synthetic Biology, Algorithmic Finance, Orbital Logistics, and Neural Bio-Interfaces

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Status: Archive-Ready / Path A Academic Extension
Governing Invariants: $C_{\text{ops}} = 0$ | $\tau_{\text{override}} \leq 11.99\text{ms}$ | Entropic Drift $(\Delta E) = 0$
Target Baseline: Phase 7 Realization (`v1.3.0-exec`)

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1 Executive Summary & Domain Generalization Axiom

ARC-SPEC-2026-EDU-01 through EDU-08 established Deterministic Admissibility Engineering (DAE) within bare-metal computing, edge networks, real-time microkernels, and mechatronic robotics. However, computing is merely a sub-domain of a larger physical reality: complex dynamic systems operating under resource constraints.

In synthetic biology, high-frequency finance, orbital aerospace, bio-neural interfaces, and automated governance, systems fail for the exact same reason software crashes: permissive, unconstrained execution. Systems initiate state changes (RNA transcription, market order fills, BCI neural stimulation) before validating structural admissibility, leading to cascading system degradation, flash crashes, cytotoxic drift, or catastrophic actuator overtravel.

This specification formalizes Universal Systems Transduction (UST) under ARC-SPEC-2026-EDU-09. It maps DAE's core invariants—5D Dynamic State Vectors ($\mathbf{\Psi}_{\text{Entity}}$), Non-Collapsible Poset Dominance Lattices, and Kingman Queue Clamping—into five major non-computer-science academic disciplines, establishing DAE as a universal, fail-closed science for interdisciplinary research.

Universal Transduction Paradigm Comparison

PERMISSIVE DOMAIN MODEL (Legacy Interdisciplinary Science):
[Trigger / Signal] $\longrightarrow$ [Unbounded Execution / State Shift] $\longrightarrow$ [Post-Hoc Failure / Crisis]

UNIVERSAL DAE TRANSDUCTION (ARC-SPEC-2026-EDU-09):

[Trigger / Signal] $\longrightarrow$ [Evaluate 5D Admissibility Gate] $\longrightarrow$ [PASS (Admissible Action)]
$\Downarrow$
$\longrightarrow$ [Short-Circuit FAIL / FREEZE] (Zero Entropic Drift)

2 Cross-Domain Invariant Transduction Mapping Matrix

To generalize DAE across non-computing fields, Table 1 formalizes the state vector, lattice failure, and queuing boundary maps for synthetic biology, finance, aerospace, neural interfaces, and law.

Table 1: Cross-Domain Invariant Transduction Mapping Matrix

Domain	**State Vector Mapping (Ψ_{Entity})**	**Poset Lattice Action (FAIL)**	**Kingman Boundary ($\rho \geq 0.95$)**
Synthetic Bio	$\mathcal{C}$ = Ribosomal Efficiency, $\mathcal{T}$ = ATP Ticks	Apoptosis / Cell Cleavage (0V)	Suppress RNA Transcription
Quant Finance	$\mathcal{A}$ = Order Size Ratio, $\mathcal{I}$ = Impact	Instant Liquidity Hold / Cancel	Wire-speed order drop (`XDP_DROP`)
Aerospace	$\mathcal{S}$ = Radiation Jitter, $\mathcal{T}$ = Clock Skew	Autonomous Attitude Hold	Downsample Telemetry Stream
Neural BCIs	$\mathcal{A}$ = Stimulus Amplitude, $\mathcal{C}$ = Impedance	Clamp Voltage to 0mV	Throttle Frame Rate under Noise
Legal Policy	$\mathcal{I}$ = Blast Radius, $\mathcal{C}$ = Statutory Bounds	Pre-Execution Refusal	Suppress Smart-Contract Trigger

3 Mathematical Generalization of Interdisciplinary Admissibility

3.1 Synthetic Biology & Genetic Circuit Admissibility

In bio-engineering, synthetic gene circuits are modeled as finite-state machines operating in cellular substrates. DAE formalizes genetic circuit admissibility:

$$\text{Admissibility}(\text{Gene_Expression}) = \begin{cases} \text{TRANSCRIPTION_PASS} & \text{if } \mathcal{C}_{\text{cellular_ATP}} \geq \text{Threshold}_{\min} \\ \text{APOPTOSIS_FREEZE} & \text{if } \dot{\mathcal{T}}_{\text{ribosome}} > \text{Drift}_{\max} \end{cases} \tag{1}$$

3.2 Quantitative Finance & Market Structural Stability

In automated financial exchanges, high-frequency trading queues behave according to Kingman's Queue Saturation dynamics. When arrival rate λ approaches clearing capacity μ ($\rho \to 1.0$), order book latency approaches infinity. EDU-09 applies Proportional Windowing Control (PWC / POSIX 12) directly to exchange ingress engines:

$$\text{Exchange_Action}(\rho) = \begin{cases} \text{FILL_ORDER (PASS)} & \text{for } \rho \leq 0.80 \\ \text{THROTTLE_CANCEL (PWC - POSIX 12)} & \text{for } 0.80 < \rho < 0.95 \\ \text{DROP_ORDER (REFUSAL - POSIX 32)} & \text{for } \rho \geq 0.95 \end{cases} \tag{2}$$

4 Interdisciplinary Research & Grant Proposal Template

To assist university faculties outside Computer Science in securing research grants using DAE, EDU-09 provides standardized interdisciplinary proposal modules.

> *"This project applies Deterministic Admissibility Engineering (DAE) (ARC-SPEC-2026-EDU-09) to brain-computer interfaces (BCIs). By implementing a bare-metal, non-collapsible dominance lattice directly on neural implant microcontrollers, we guarantee that AI-decoded motor intentions cannot exceed safe tissue voltage caps. The system evaluates neural stimulus state vectors (Ψ_{Neural}) in sub-12ms execution windows ($\tau_{override} \leq 11.99ms$), providing hardware-enforced patient safety."*

SPECIFICATION COMPLETE: ARC-SPEC-2026-EDU-09
DEPLOYMENT POSTURE: ARCHIVED FOR FUTURE PATH A EXTENSION
AUTHORIZATION HASH: A-77-DELTA-SHIELD-LOCKED